1-to'plam. *Soddalashtiring – avval ko'paytuvchilarga ajrating, keyin faqat ko'paytuvchilarni qisqartiring.*

1. $\dfrac{x^2-9}{x+3}=$

2. $\dfrac{x^2+5x+6}{x+2}=$

3. $\dfrac{x^2-5x+6}{x^2-4}=$

4. $\dfrac{2x^2+4x}{x^2-4}=$

5. $\dfrac{x^2-1}{x^2+2x+1}=$

6. $\dfrac{x^2-7x+10}{x^2-25}=$

2-to'plam. *Ko'paytiring yoki bo'ling.*

1. $\dfrac{x+4}{x-3}\times\dfrac{x-3}{x+1}=$

2. $\dfrac{x^2-16}{3x}\times\dfrac{x}{x+4}=$

3. $\dfrac{x-2}{x+5}\times\dfrac{x+5}{x-2}=$

4. $\dfrac{x^2-9}{x+2}\div\dfrac{x+3}{x+2}=$

5. $\dfrac{x+1}{x-5}\div\dfrac{x+1}{2x-10}=$

6. $\dfrac{x^2-4}{5}\div\dfrac{x-2}{10}=$

3-to'plam. *Tenglamani yeching (hali tuzoq yo'q).*

1. $\frac{x}{3} + 1 = \frac{x+4}{6}$

2. $\frac{2}{x} + 1 = 3$

3. $\frac{x+1}{x+4} = \frac{1}{2}$

4. $\frac{3}{x-1} = \frac{2}{x+2}$

5. $\frac{x}{5} + \frac{1}{5} = \frac{6}{x}$ *(ikkita javob)*

4-to'plam. *Yeching va TEKSHIRING – ba'zi nomzodlar rad etilishi, ba'zilari esa qabul qilinishi kerak. Javobni son yoki "yechim yo'q" deb aniq ko'rsating.*

1. $\frac{x}{x-3} = \frac{3}{x-3} - 2$

2. $\frac{x+1}{x-4} = \frac{5}{x-4}$

3. $\frac{2x}{x-5} = \frac{10}{x-5} + 1$

4. $\frac{x+2}{x-1} = 3$ *(bu tuzoq emas – tekshiring)*

5-to’plam. *Aralash – har bir band o’z ko’rsatmasiga ega.*

1 Soddalashtiring: $\dfrac{2x^2-8}{x^2-4}=$

2 Yeching: $\dfrac{4}{x+1}=\dfrac{2}{x-1}$

3 Ko’paytiring: $\dfrac{x-5}{x+3}\times\dfrac{x+3}{2x}=$

4 Yeching va tekshiring: $\dfrac{1}{x-2}+\dfrac{1}{x+2}=\dfrac{4}{x^2-4}$

5 Soddalashtiring: $\dfrac{x^2-6x+9}{x^2-9}=$

6-to’plam. *Xatoni toping – buzilgan qadamni ayting, to’g’ri javobni bering.*

1 $\dfrac{x+5}{x}$ ni soddalashtiring – ko’rsatilgan yechimda xato bor:

“x’larni qisqartiramiz: javob 5.”

Xato va to’g’ri javob: .....

2 $\dfrac{x^2-4}{x+2}$ ni soddalashtiring – ko’rsatilgan yechimda xato bor:

“Hammasi qisqaradi, javob 0.”

Xato va to’g’ri javob: .....

3 $\dfrac{3x}{x-2}=\dfrac{6}{x-2}+x$ ni yeching – ko’rsatilgan yechimda xato bor:

“Tozalab, $x^2-5x+6=0$*, demak* $x=2$ *yoki* $x=3$ *– ikkalasi ham yechim.”*

Xato va to’g’ri javob: .....

4 $\dfrac{3x+6}{3x}$ ni soddalashtiring – ko’rsatilgan yechimda xato bor:

“3’larni va x’larni qisqartiramiz: javob 6.”

Xato va to’g’ri javob: .....

6-to'plam (davomi).

5 $\frac{x-2}{x+3}\times\frac{x+3}{x-2}$ ni ko'paytiring – ko'rsatilgan yechimda xato bor:

"Hammasi qisqaradi, javob 0*."*

Xato va to'g'ri javob: .....

7-to'plam. *Bonus / qiyin.*

1 $\frac{2}{x-1}-\frac{3}{x+1}=1$ *(ikkita javob)*

2 $\frac{x^3-x}{x^2-1}=$

3 $\frac{x+1}{x-3}=\frac{x-1}{x+3}$

4 $\frac{1}{x}+\frac{1}{x+2}=\frac{3}{4}$ *(ikkita javob)*

Javoblar

To'plam	#	Javob	#	Javob
1-to'plam	1	$x-3$	2	$x+3$
	3	$\frac{x-3}{x+2}$	4	$\frac{2x}{x-2}$
	5	$\frac{x-1}{x+1}$	6	$\frac{x-2}{x+5}$
2-to'plam	1	$\frac{x+4}{x+1}$	2	$\frac{x-4}{3}$
	3	1	4	$x-3$
	5	2	6	$2x+4$
3-to'plam	1	$x=-2$	2	$x=1$
	3	$x=2$	4	$x=-8$
	5	$x=-6$ yoki $x=5$		
4-to'plam	1. $x=3$ emas – mahraj $x-3$ nolga aylanadi. **Yechim yo'q.**			
	2. $x=4$ emas – mahraj $x-4$ nolga aylanadi. **Yechim yo'q.**			
	3. $x=5$ emas – mahraj $x-5$ nolga aylanadi. **Yechim yo'q.**			
	4. $x=\frac{5}{2}$ – mahraj $\frac{3}{2}\neq 0$, **to'g'ri yechim** (tuzoq emas).			
5-to'plam	1	2	2	$x=3$
	3	$\frac{x-5}{2x}$	4	yechim yo'q
	5	$\frac{x-3}{x+3}$		
6-to'plam	1. Xato – x surat $(x+5)$ ning hadi, ko'paytuvchisi emas. To'g'ri: $\frac{x+5}{x}$ allaqachon soddalashtirilgan.			
	2. Xato – $(x+2)$ qisqargach $(x-2)$ qoladi, yo'qolmaydi. To'g'ri: $x-2$.			
	3. Xato – $x=2$ asl mahrajda tekshirilmagan, u nolga aylanadi. To'g'ri: faqat $x=3$.			
	4. Xato – x faqat bitta hadning ko'paytuvchisi. To'g'ri: $\frac{3x+6}{3x}=\frac{x+2}{x}$, ortiq soddalashmaydi.			
	5. Xato – ko'paytuvchilar qisqaradi, lekin natija 1 bo'ladi, 0 emas. To'g'ri: 1.			
7-to'plam	1	$x=-3$ yoki 2	2	x
	3	$x=0$	4	$x=-\frac{4}{3}$ yoki 2

Video yechimlar va qo'shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

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